Massive activations: refers to a small number of specific neurons that consistently output values that are orders of magnitude (100× or 1000×) larger than all other neurons

→ mostly observed in dense layer: they tend to appear in specific features dimensions and stay massive regardless of the input text

Problem: quantization difficulty → related to model compression

→ INT8 (8-bit) → we map all floating-point numbers to 0-255

Say we have weights [0.01, 0.21, ..., 1000.02]

→ to store this large value, we squeeze 0.01 → 0. decrease model's accuracy.

Solution: Compressing them seperately

possible explanation: they act as bias terms that shift the model's focus

$Ax+b$

$\boxed{\text{Attention sink}}$: refers to a phenomenon where the model allocates a massive amount of attention to the very first token <SOS>, even though this token has no semantic meaning for the current context.

$\boxed{\text{Problem}}$: Say we consider StreamingLLM where we want the model to output infinite length of text. We then use a sliding window strategy

(because the computational time is n^2 for paying attention to all words before)

This strategy will fail because the window does not include the <SOS> token, which other tokens often pay a lot of attentions to.

$\boxed{\text{Solution}}$: keep the first few tokens while moving the window.

$\boxed{\text{Possible explanation}}$: When token does not need to pay much attention to other tokens, it just dump the other attention to <SOS>

→ a common token for all tokens.

Check gated attention for relieving this problem as well.